

EDUCATION**Peking University, Master of Public Health, Epidemiology and Biostatistics****Sep 2025 – Present**

- GPA: 3.49/4.00
- Supervisor: Prof. Yuantao Hao
- Key Courses: Advanced Statistics, Machine Learning, Python Programming

Expected graduation: Jun 2027**Peking University, Bachelor of Medicine, Preventive Medicine****Sep 2020 – Jun 2025****Peking University, Bachelor of Science, Mathematics and Applied Mathematics (Dual Degree)****Sep 2021 – Jun 2025**

- Overall GPA: 3.30/4.00 (82.75/100.00)
- Key Courses: Epidemiology, Health Statistics, Advanced Algebra, Mathematical Analysis, Probability Theory
- Award: Outstanding Undergraduate Thesis Award, 2025

RESEARCH INTERESTS

Infectious disease dynamics; Infectious disease modelling; Machine learning; Epidemic forecasting

RESEARCH EXPERIENCE**Influenza Dynamic Identification and Forecasting with Physics-Informed Neural Networks****May 2026 – Present***Master's Thesis, Peking University*

- **Dynamic Identification:** Developing a SIRS-constrained physics-informed neural network to estimate latent epidemic states and time-varying dynamics from influenza surveillance data.
- **Epidemic Forecasting:** Extending the inferred epidemic dynamics to influenza and subtype-specific forecasting.
- **Model Validation:** Evaluating parameter recovery and forecast performance using simulated epidemics and real-world surveillance data, with benchmark-model comparisons and uncertainty assessment.

Chikungunya Transmission Chain Reconstruction and Serial Interval Estimation**Nov 2025 – Jan 2026***Research Contributor, Project for Chikungunya Outbreak in China, Peking University*

- **Data Integration:** Analyzed epidemiological data from 1,329 reported cases together with available genomic sequences from the 2025 Guangzhou Chikungunya outbreak.
- **Transmission Inference:** Applied *outbreaker2* to infer probable transmission links using symptom-onset dates and genomic information and fitted gamma distributions to characterize the serial interval distribution.
- **Quantitative Analysis:** Led the coding, statistical analysis, visualization, and drafting of the corresponding methodological and results sections.

Epidemiological Interaction Between Influenza Virus and SARS-CoV-2**Jan 2025 – May 2025***Undergraduate Thesis, Peking University*

- **Mechanistic Model:** Developed a coupled SIR–SEIR model to investigate bidirectional interactions between influenza and SARS-CoV-2 using Hong Kong surveillance data.
- **Parameter Estimation:** Estimated transmission and interaction parameters using maximum likelihood estimation, with bootstrap resampling to quantify uncertainty.
- **Research Output:** Produced two first-author studies; one accepted and one under review.

PUBLICATIONS

* Equal contribution

- **Qin, Z.**, Wei, Y., Zhou, J., Du, Z., Zhao, S., Peng, Z., & Hao, Y. Bidirectional synergistic interaction between influenza virus and SARS-CoV-2 in the post-pandemic era: A modelling study. *Influenza and Other Respiratory Viruses*. Accepted.
- Wang, L., Wang, Q., **Qin, Z.**, Zheng, F., Zhang, X., & Yu, T. (2026). How organizational commitment shapes IPC execution through job satisfaction among infection prevention and control professionals. *Frontiers in Public Health*, 14, 1899025.
- Wang, Q. *, **Qin, Z. ***, Qin, L., Xing, X., Wang, Z., Cui, H., Zhang, J., Xing, Y., Chen, J., Zhao, M., Luo, W., Zhu, H., & Jia, Z. (2026). Distribution patterns of narcotic drugs, psychotropic substances, and new psychoactive substances among reported drug users in China (2014-2026): A systematic review and meta-analysis. *Addictive Behaviors Reports*, 23, 100688.

- Yang, Y., Xing, X., **Qin, Z.**, Han, C., & Zhu, H. (2026). Alcohol consumption and age-specific risk of esophageal cancer: Prospective cohort study. *JMIR Public Health and Surveillance*, 12, e92949.
- Liu, J.*, **Qin, Z.***, Liu, R., Luo, W., Jia, Z., Wu, L., & Wu, J. (2026). A study of early-warning model for influenza surveillance in hospitalized patients in three hospitals in Guangdong. *Disease Surveillance*, 41(1), 74-79. (in Chinese)
- Liu, J., **Qin, Z.**, Xi, Z., Zhou, Y., & Xu, Y. (2025). Effect of dietary patterns on maternal body composition and bone mineral density during three trimesters in Chinese pregnant women: A cross-sectional study. *Nutrients*, 17(12), 2021.
- Yang, Y., Han, C., Xing, X., **Qin, Z.**, Wang, Q., Lan, L., & Zhu, H. (2025). Effects of postoperative complications on overall survival following esophagectomy: A meta-analysis using the restricted mean survival time analysis. *Thoracic Cancer*, 16(3), e70011.
- Chen, J.*, **Qin, Z.***, & Jia, Z. (2024). The application status of sequencing technology in global respiratory infectious disease diagnosis. *Infection*, 52, 2169–2181.
- **Qin, Z.**, Xi, Z., Liu, J., Hu, Y., Tan, F., & Jia, Z. (2024). Legislation of infectious disease laws in major international organizations, unions and countries in the world. *Disease Surveillance*, 39(7), 881-887. (in Chinese)
- Wang, Q.*, **Qin, Z.***, Xing, X., Zhu, H., & Jia, Z. (2024). Prevalence of cannabis use around the world: A systematic review and meta-analysis, 2000-2024. *China CDC Weekly*, 6(25), 597-604.

MANUSCRIPTS UNDER REVIEW

- **Qin, Z.**, Han, X., Wei, Y., Zhao, S., Hao, Y. From Importation Prevention to Outbreak Control at the Source: The Role of Non-Endemic Countries in the Bundibugyo Ebola Outbreak.
- **Qin, Z.**, Wei, Y., & Hao, Y. Influenza and COVID-19 activity in post-pandemic Hong Kong, 2023-2026: An ecological surveillance study.
- Ma, Y., Liu, Y., Yuan, J., Guo, Z., Nama, N., Fan, S., Ma, Q., Zhen, R., Sun, X., Zhou, J., Zhou, J., Su, W., **Qin, Z.**, Luo, L., Zhao, Z., Li, K., Chi, Q., Wu, X., Wang, Y., Yang, Y., Chong, M. K. C., Ali, S. T., He, D., Zhao, S., Hao, Y., & Zhang, Z. Epidemiology of chikungunya virus outbreak in Guangzhou, China in 2025: a population-based observational study.
- **Qin, Z.**, Wang, Y., Xing, X., Hu, Z., Feng, H., & Zhu, H. Mapping polydrug use network and racial/ethnic heterogeneity among adolescent drug users in the United States.

ACADEMIC SERVICE

- Assisted with peer review under faculty supervision for Biomedical and Environmental Sciences, Chinese Journal of Preventive Medicine, and PLOS Computational Biology
- Assisted with proofreading for the 9th edition of Health Statistics, People's Medical Publishing House
- Contributed to continuing education content for the Public Health Workforce Development chapter of the 2025 China Public Health Development Report

INTERNSHIP

Clinical Intern, Peking University Third Hospital Haidian Campus

Feb 2023 – Jan 2024

- Completed full-time clinical rotations across major departments including internal medicine, surgery, obstetrics and gynecology, and pediatrics
- Participated in outpatient care, ward rounds, operations, basic examinations, and medical record writing

Public Health Intern, Beijing Xicheng District Health Commission and Xicheng District CDC

Jul 2024 – Sep 2024

- Completed full-time rotations in the Medical Administration Department of the Health Commission and CDC departments including laboratory testing, immunization planning, and health education.
- Participated in data work, document writing, routine administrative analysis, and meeting support.

HONORS AND AWARDS

- Outstanding Undergraduate Thesis Award, Peking University, 2025.
- Third Prize, Guangdong Provincial Infectious Disease Surveillance and Early-Warning Modelling Competition, 2025.
- First Prize, Peking University Challenge Cup, 2023.
- Second Prize, National College Student Statistical Modeling Competition, 2022.
- Third Prize, National College Student Mathematics Competition, 2022.
- Third Prize, Mathematical Modeling and Computer Applications Competition, Peking University, 2022.

SKILLS

Coding & Research Skills: Python, R, Latex, Zotero, Endnote

Languages: Chinese (native), English (IELTS planned)